

Wojciech Trejter

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Summary

AI engineer and researcher working where machine learning meets the physical world — robotic perception, SLAM, computer vision, and world models that transfer from simulation to reality, with published work on faithfulness and interpretability. MSc Artificial Intelligence (Cum Laude, University of Amsterdam), author in *Transactions on Machine Learning Research*, winner of the European Rover Challenge 2023, two-time ICPC competitor.

Work History

AI Intern — **xBerry** — *May – Aug 2026, Wroclaw*

- Built a sim-to-real transfer pipeline between Isaac Lab simulation and real robotic hardware; the team raised real-robot task success from 0% to 80%.
- Fine-tuned world models with LoRA (Low-Rank Adaptation), adapting large models within a limited compute budget.
- Architected simulation pipelines for training and evaluating high-fidelity robotic arm tasks in Isaac Lab.

Teaching Assistant — **University of Amsterdam** — *Sep 2025 – Apr 2026, Amsterdam*

- Led seminars and tutorials for the Computer Vision and Reinforcement Learning courses, turning abstract concepts (convolutional filters, policy gradients) into example-driven explanations.
- Graded coursework and mentored students on assignments, targeting feedback at commonly misunderstood theory.

Software Engineer Intern — **IBM CIC** — *Feb 2024 – Jun 2024, Groningen*

- Delivered an internal full-stack dashboard (MongoDB, Express, React, Node — MERN) end-to-end, from database schema to UI.
- Worked in a Scrum team on bi-weekly sprints, running standups and sprint reviews.

Technical Team Lead — **Makercie** — *Jan 2023 – Jun 2024, Groningen*

- Took **1st place** at the European Rover Challenge Remote Edition (2023), leading a 7-person software team (Jan – Sep 2023); then team member until Jun 2024.
- Built the rover's autonomy stack in Robot Operating System (ROS): computer-vision-driven control of a 7-DoF robotic arm.

Education

MSc Artificial Intelligence — **University of Amsterdam** — *Sep 2024 – Aug 2026, Amsterdam*

- **Cum Laude** with GPA 8.3/10.
- **Thesis**: *Evo-SLAM: Monocular Gaussian Splatting SLAM for Evolving Scenes*, an online system built on Vision Transformers and foundation models, outperforming published state-of-the-art methods (GaME, WildGS-SLAM); paper under review.
- Courses: Computer Vision 1&2, Deep Learning 1&2, Machine Learning 1&2, NLP, MLOps.

BSc Artificial Intelligence — **University of Groningen** — *Sep 2021 – Jul 2024, Groningen*

- **Cum Laude** with GPA 8.8/10 and 270 ECTS (180 core + 45 Honours + 45 extracurricular).
- Finished with Honours; Honours project on deep-learning image classification for Portable Antiquities Netherlands (see Projects).
- **Thesis**: *Application of Multitask Learning Paradigm on Hydrodynamic Imaging*.
- Courses: Probability Theory, Neural Networks, Computer Graphics, Computational Complexity.

Publications

- Kurek, I., **Trejter, W.**, Frkovic, S., Erdelez, A., [Re] Improving Interpretation Faithfulness for Vision Transformers, *Transactions on Machine Learning Research 2025*.
  - Reproduced a published method for faithful Vision Transformer explanations; the results did not support the original paper's claims.
  - Ran experiments on an HPC cluster (SLURM).
- **Trejter, W.**, Wolf, B. J., Application of Multitask Learning Paradigm on Hydrodynamic Imaging, BSc thesis.
  - Multitask deep learning (MLP, LSTM) on flow-sensor data for object localisation and shape classification.

Projects

- **SAR Ship Locator** (2026): Weekend project to learn SAR basics: Sentinel-1 ship detection checked against AIS.
- **GAPC** (Groningen Algorithmic Programming Contest, 2023): Organised the contest and designed its problem set.
- **Image Classification for Antiquities**: Honours project: image classification by fine-tuning Residual Networks (ResNets) on a dataset obtained from Portable Antiquities Netherlands.
- **Rodina AI Startup**: Co-founder and programmer for a B2B startup creating synthetic image datasets; built and delivered an MVP to a client.

Awards

- **European Rover Challenge Remote Edition (2023)**: 1st place as Technical Lead (see Makercie).
- **International Collegiate Programming Contest (2022, 2023)**: 28th in Benelux, 58th in Northwestern Europe.
- **Beta Business Days Startup Competition (2024)**: 1st place, following a rapid development of a viable business strategy.

Skills

Programming: Python, C, Java, SQL, Bash, Linux
Machine Learning: PyTorch, Keras, scikit-learn, Hugging Face Transformers, LoRA fine-tuning, LLMs, BERT, Diffusion Models, Reinforcement Learning, NLP, Explainable AI
Computer Vision & Robotics: SLAM, 3D Gaussian Splatting, Structure from Motion, OpenCV, ROS, Isaac Lab, Isaac Gym, Sim-to-Real, World Models
Data & Infrastructure: NumPy, Pandas, Matplotlib, Parquet, Docker, FastAPI, SLURM/HPC, WandB, MongoDB, React, Node.js
Engineering Practice: Git, GitLab CI/CD, Unit Testing, Agile/Scrum

Languages

Polish: Native
English: Full Professional Proficiency

Volunteering

Volunteer Firefighter **Poznan** *2017 – 2021*

Emergency Response Officer **Cover Student Association** *2022 – 2024*